Literature Survey on Multimodal Sentiment Analysis

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1. Introduction

Sentiment analysis has evolved from rudimentary text classification into a sophisticated discipline capable of interpreting the nuanced spectrum of human affective states. Traditional unimodal approaches, confined to lexical analysis of written content, proved fundamentally limited when confronted with the multimodal nature of human communication—where linguistic content, vocal prosody, facial expressions, and physiological signals interact synergistically to convey emotional meaning [1], [2]. This limitation became particularly pronounced in digital environments dominated by rich media content on platforms such as YouTube, TikTok, and Instagram, where users routinely express sentiment through heterogeneous data streams that cannot be adequately interpreted through text alone [3], [4].

Multimodal Sentiment Analysis (MSA) emerged as a paradigm shift that integrates natural language processing, computer vision, and speech processing to construct a holistic representation of emotional states [5], [6]. By synthesizing information across modalities—including textual semantics, acoustic prosody (pitch, rhythm, intensity), visual cues (facial action units, gestures), and increasingly physiological biomarkers (EEG, heart rate variability)—MSA systems approximate human cognitive processes of sensory integration [7], [8]. This multimodal integration proves essential for resolving ambiguities inherent in unimodal analysis, such as detecting sarcasm when positive lexical content contradicts frustrated vocal tone or facial micro-expressions [9], [10].

The field has matured through distinct methodological phases: from early feature concatenation approaches to sophisticated attention-driven fusion architectures, and most recently to reasoning-capable Multimodal Large Language Models (MLLMs) that generate interpretable sentiment rationales [11], [12]. Standardized benchmark datasets—including CMU-MOSI (2,199 video segments), CMU-MOSEI (23,453 utterances), MELD (13,000+ conversational utterances), and IEMOCAP (dyadic interactions)—have enabled systematic evaluation using metrics such as binary accuracy (Acc-2), seven-class accuracy (Acc-7), F1-score, mean absolute error (MAE), and Pearson correlation [13], [14]. These benchmarks reveal consistent performance improvements: early fusion models achieved approximately 77.6% Acc-2 on CMU-MOSI, whereas contemporary attention-based architectures like the Multimodal Transformer (MulT) reach 81.1%, with state-of-the-art contrastive learning approaches (e.g., DCF) attaining 86.62% [15], [16].

This survey synthesizes advancements in MSA methodology between 2019 and 2025, critically examining the evolution of representation learning, fusion strategies, and architectural paradigms. We analyse the transition from handcrafted feature engineering to self-supervised representation learning, evaluate the trade-offs between fusion methodologies, and assess the transformative impact of large language models on sentiment reasoning capabilities. Crucially, we identify persistent challenges—including modality misalignment, missing data robustness, demographic bias, and computational constraints—that define the frontier of MSA research. The scope encompasses technical architectures, dataset evolution, evaluation methodologies, and emerging applications in healthcare, social media analytics, and human-robot interaction, while deliberately excluding peripheral topics such as pure emotion recognition without sentiment polarity or unimodal sentiment analysis techniques.

2. Foundations of Multimodal Sentiment Analysis

2.1 Modalities and Feature Representation

MSA systems typically process four primary modalities, each requiring specialized feature extraction pipelines to transform raw signals into semantically meaningful embeddings. *Textual modality* remains the semantic anchor for sentiment interpretation, having evolved from static word embeddings (Word2Vec, GloVe) to context-aware representations generated by Transformer-based pre-trained language models (PLMs) such as BERT and RoBERTa [17], [18]. These models capture bidirectional dependencies and long-range contextual relationships through self-attention mechanisms, enabling accurate interpretation of sentiment-bearing words whose polarity depends on syntactic context [19]. Recent advances incorporate dependency-syntax-guided pruning to focus attention on words most relevant to specific sentiment targets in aspect-based analysis [20].

Acoustic modality provides critical prosodic context that text alone cannot convey. Traditional feature extraction relies on low-level descriptors (LLDs) including Mel-frequency cepstral coefficients (MFCCs), pitch contours, and intensity dynamics, typically extracted using tools like OpenSMILE [21], [22]. Modern approaches leverage self-supervised audio representation learning (e.g., Wav2Vec 2.0, Data2Vec Audio) that learns directly from raw waveforms, capturing nuanced vocal modulations even in noisy environments [23]. Architectures like AViSE-Net employ dilated convolutional neural networks to model long-range temporal dependencies in emotional prosody evolution across utterances [24].

Visual modality captures immediate non-verbal emotional markers through facial expressions, gestures, and scene context. Early approaches utilized handcrafted features like Local Binary Patterns (LBP) and Histogram of Oriented Gradients (HOG), but contemporary systems employ deep convolutional architectures (ResNet-50, EfficientNet, VGG19) to extract high-dimensional feature vectors [25], [26]. Specialized tools like OpenFace detect facial landmarks and estimate Facial Action Units (AUs)—standardized descriptors of muscle movements associated with basic emotions [27]. Recent research emphasizes that scene-level semantics (e.g., environmental context) significantly influence emotional perception beyond facial expressions alone [28].

Physiological modality represents an emerging frontier incorporating objective biomarkers less susceptible to conscious manipulation than behavioral signals. Electroencephalogram (EEG) data (e.g., from DEAP dataset recorded at 512 Hz) and heart rate variability (HRV) extracted via remote photoplethysmography (rPPG) from facial video provide internal state markers valuable for clinical applications [29], [30]. These signals are annotated along dimensional affect frameworks (Valence-Arousal-Dominance) rather than discrete categories, enabling fine-grained emotional assessment [31].

2.2 Fusion Strategies: From Concatenation to Cross-Modal Reasoning

The integration of heterogeneous modalities constitutes the core technical challenge in MSA. Fusion strategies have evolved through distinct generations reflecting increasing sophistication in modeling cross-modal interactions.

Early fusion (feature-level) concatenates unimodal feature vectors into a single high-dimensional representation before classification [32]. While enabling models to learn low-level cross-modal correlations from inception, this approach suffers from the "curse of dimensionality" and synchronization challenges when modalities operate at different sampling rates (e.g., audio at kHz versus video at fps) [33]. *Late fusion* (decision-level) processes each modality through independent classifiers and aggregates outputs via weighted averaging or voting [34]. This strategy demonstrates robustness to missing modalities but fails to capture intricate feature-level interactions essential for nuanced sentiment interpretation [35].

Tensor-based fusion addresses these limitations by explicitly modeling higher-order interactions. The Tensor Fusion Network (TFN) computes outer products of unimodal features to capture unimodal, bimodal, and trimodal relationships simultaneously [36]. However, computational complexity grows exponentially with modality count. Low-rank Multimodal Fusion (LMF) mitigates this through tensor decomposition, reducing complexity to linear while preserving interaction modeling capability [37]. Recent innovations like Multimodal Mixture of Low-Rank Experts (MMoLRE) achieve up to 80% parameter reduction compared to standard Mixture of Experts while maintaining competitive performance [38].

Attention-based fusion represents the current state-of-the-art, enabling dynamic weighting of modalities based on contextual relevance. Cross-modal attention mechanisms allow one modality to query information from another (e.g., text modality attending to acoustic features when vocal stress aligns with specific words), effectively aligning features in a joint latent space without explicit pre-alignment [39], [40]. The Multimodal Transformer (MulT) implements directional pairwise cross-modal attention that latently adapts asynchronous sequences, achieving 81.1% Acc-2 on CMU-MOSI without Connectionist Temporal Classification alignment modules required by earlier architectures [41]. Dynamic Attention Fusion (DAF) further optimizes this paradigm through lightweight adaptive weighting per utterance without fine-tuning underlying encoders, enhancing deployability in resource-constrained environments [42].

2.3 Benchmark Datasets and Evaluation Methodology

Standardized datasets provide the foundation for comparative assessment and methodological progress in MSA. These corpora have evolved from small curated collections to large-scale repositories reflecting real-world complexity and cultural diversity.

Table 1: Comparative Overview of Core MSA Benchmark Datasets

Dataset	**Modalities**	**Size (Segments)**	**Annotation Scheme**	**Key Characteristics**
CMU-MOSI	Text, Audio, Video	2,199	Sentiment intensity (-3 to +3)	YouTube opinion videos; foundational benchmark
CMU-MOSEI	Text, Audio, Video	23,453	Sentiment intensity & 7 emotions	Largest English corpus; 1,000+ speakers
MELD	Text, Audio, Video	13,000+	7 discrete emotions	Multi-party dialogue from *Friends* TV series
IEMOCAP	Text, Audio, Video	10,000+	9 emotions + dimensional	Dyadic conversational interactions
CH-SIMS v2	Text, Audio, Video	14,563	Fine-grained non-verbal cues	Chinese language context; cultural specificity
MVSA-Single	Text, Image	4,869	Polarity (positive/negative/neutral)	Twitter image-text pairs; aspect-level focus

Evaluation metrics reflect the dual nature of sentiment tasks: classification (discrete polarity/emotion categories) and regression (continuous intensity dimensions). Binary accuracy (Acc-2) and seven-class accuracy (Acc-7) measure classification performance, while F1-score addresses class imbalance common in sentiment distributions [43]. For regression tasks, mean absolute error (MAE) quantifies deviation from ground-truth intensity values, and Pearson correlation (Corr) measures linear relationship strength between predicted and actual values [44]. Recent benchmarks demonstrate consistent performance stratification: models employing contrastive learning (DCF: 86.62% Acc-2 on CMU-MOSI) and attention-guided tensor fusion (BIMHA) outperform traditional concatenation approaches by 5–9 percentage points [45], [46].

3. Architectural Paradigms in Multimodal Sentiment Analysis

3.1 Classical Machine Learning Foundations

Early MSA systems adapted unimodal sentiment techniques by concatenating handcrafted features from each modality into a unified vector processed by classical classifiers. Support Vector Machines (SVMs) demonstrated superiority over Naive Bayes for complex review data (78% versus 76% accuracy on IMDb), leveraging kernel methods to handle non-linear decision boundaries in high-dimensional feature spaces [47]. Random Forests provided robustness to noisy features through ensemble averaging but struggled with temporal dependencies inherent in sequential modalities like speech and video [48]. These approaches required extensive domain expertise for feature engineering and proved brittle when confronted with distribution shifts between training and deployment environments—a limitation that catalyzed the transition to representation learning paradigms.

3.2 Deep Learning Architectures and Temporal Modeling

The advent of deep learning enabled end-to-end learning of modality-specific representations. Early multimodal systems combined Convolutional Neural Networks (CNNs) for visual feature extraction with Long Short-Term Memory (LSTM) networks for sequential modeling of text and audio [49]. Bidirectional LSTMs (BiLSTMs) captured context from both past and future time steps, improving detection of sentiment shifts within utterances [50]. Gated Recurrent Units (GRUs) offered computational efficiency through simplified gating mechanisms while maintaining competitive performance on temporal modelling tasks [51].

Architectural innovation focused on addressing temporal misalignment between modalities. While audio signals convey emotional prosody continuously throughout an utterance, visual cues like facial expressions may peak at specific moments, and textual semantics anchor to discrete lexical units [52]. Hybrid architectures incorporating modality-specific encoders followed by cross-modal attention layers proved essential for modelling these asynchronous interactions without explicit temporal alignment—a requirement that significantly complicated pipeline design in earlier systems [53].

3.3 Transformer-Based Multimodal Integration

Transformer architectures revolutionized MSA by replacing recurrent temporal modelling with self-attention mechanisms capable of capturing long-range dependencies regardless of sequence position [54]. The Multimodal Transformer (MulT) specifically addressed the non-alignment challenge through directional pairwise cross-modal attention that operates directly on unaligned sequences [55]. Mathematically, cross-modal attention computes:

$$\mathbf{Attention}(Q_L, K_A, V_A) = \mathrm{softmax}\left(\frac{Q_L K_A^T}{\sqrt{d_k}}\right) V_A$$

where linguistic features serve as queries (Q_L) attending to acoustic keys ($K^T{}_A$) and values (V_A), enabling dynamic selection of relevant prosodic features for each textual context [56]. This approach eliminated the need for Connectionist Temporal Classification alignment modules, reducing pipeline complexity while improving performance on unaligned benchmarks by 3.5 percentage points [57].

3.4 Graph-Based and Contrastive Learning Approaches

Conversational sentiment analysis introduced additional complexity through speaker dependencies and dialogue flow. Graph Neural Networks (GNNs) model utterances and speakers as nodes in a graph, capturing contextual dependencies through message passing between connected nodes [58]. The Three-modal Joint Representation (TMJR) model implements directional interaction paths between modalities as graph edges, effectively bridging semantic gaps in unaligned cross-media data where sequential models fail [59].

Contrastive learning emerged as a powerful paradigm for learning robust multimodal representations by pulling together embeddings of the same utterance across modalities while pushing apart embeddings of different utterances [60]. The Dual Contrastive Fusion (DCF) framework achieves state-of-the-art results (86.62% Acc-2 on CMU-MOSI) by jointly optimizing intra-modal consistency and inter-modal alignment through contrastive loss functions [61]. This approach proves particularly effective for handling noisy or incomplete modalities by learning representations that maintain semantic coherence even when individual channels are degraded.

Table 2: Comparative Analysis of MSA Architectural Paradigms

Model Type	**Fusion Strategy**	**Strengths**	**Limitations**	**Representative Dataset**
SVM + Handcrafted Features	Early fusion (concatenation)	Interpretability; low computational cost	Poor temporal modeling; feature engineering burden	Early MOSI subsets
CNN-LSTM Hybrids	Hierarchical fusion	Captures spatial + temporal patterns	Struggles with modality misalignment	IEMOCAP
Tensor Fusion Network (TFN)	Tensor product	Models high-order interactions explicitly	Exponential parameter growth; computationally intensive	CMU-MOSI
Multimodal Transformer (MulT)	Cross-modal attention	Handles unaligned sequences; end-to-end	High memory requirements for long sequences	CMU-MOSEI
DCF (Dual Contrastive Fusion)	Contrastive learning	Robust to noise; strong generalization	Requires careful temperature tuning	CMU-MOSI/MOSEI
MLLM-Guided Frameworks	Reasoning distillation	Explainable predictions; zero-shot capability	Extreme computational cost; deployment challenges	Twitter-15/17

3.5 Large Language Models and Reasoning-Capable Systems

The integration of Large Language Models (LLMs) and Multimodal Large Language Models (MLLMs) represents the most significant paradigm shift in recent MSA research, transitioning the field from discriminative classification to generative reasoning [62]. Unlike traditional models mapping inputs to closed label sets, MLLM-based frameworks like Emotion-LLaMA and GPT-4V generate natural language explanations for sentiment predictions through Chain-of-Thought (CoT) prompting [63]. This "explainable MSA" traces sentiment sources by articulating identified cues across modalities before reaching final polarity judgments—a capability critical for high-stakes domains like clinical mental health assessment [64].

The LRSA (Large-Rationale Small-Model) framework addresses computational constraints through knowledge distillation: a powerful teacher MLLM generates detailed sentiment rationales that guide training of a smaller, task-specific student model [65]. Through reasoning distillation, the student learns to mimic the teacher's predictive logic, achieving state-of-the-art results with a fraction of the parameter count [66]. Uncertainty-aware collaborative systems (U-ACS) further optimize resource allocation through sample-aware cascades where lightweight models process routine samples while reserving MLLM analytical power for ambiguous or out-of-distribution cases [67]. This architecture enables real-time responsiveness (sub-second latency) while maintaining high performance ceilings—essential for applications like live public opinion monitoring.

4. Challenges, Limitations, and Research Gaps

Despite substantial progress, MSA systems face persistent challenges that limit real-world deployment and equitable performance across diverse populations. *Modality misalignment* remains fundamentally problematic: in natural interactions, emotional cues manifest asynchronously across channels (e.g., facial expression changes preceding vocalized sentiment by 300–500ms), yet most fusion architectures assume temporal synchronization or require explicit alignment modules that introduce error propagation [68]. While MulT's cross-modal attention partially addresses this through latent alignment, performance degrades significantly under extreme asynchrony (>1 second offsets), revealing incomplete solutions to this core challenge.

Missing modality robustness constitutes another critical limitation. Real-world deployments encounter sensor failures, poor lighting conditions, or network latency causing intermittent modality loss [69]. Most fusion architectures assume consistent availability of all channels; accuracy drops precipitously (e.g., from 80% to 54% on CMU-MOSI) when trained multimodal models process unimodal inputs [70]. Recent approaches like Language-dominated Noise-resistant Learning Network (LNLN) leverage text as a "dominant modality" through adversarial learning to maintain representation quality during auxiliary modality loss [71]. However, this reinforces *text-predominance bias*—a phenomenon where models over-rely on linguistic features while underutilizing valuable non-verbal cues, particularly problematic for sarcasm detection where non-verbal channels often contradict textual content [72].

Dataset bias presents severe ethical and performance implications. Training corpora predominantly feature "WEIRD" populations (Western, Educated, Industrialized, Rich, Democratic), resulting in significantly lower accuracy for emotional expressions from underrepresented ethnicities, age groups, and neurodiverse individuals [73]. Models trained on normative expression patterns systematically misinterpret atypical expressions common in Autism Spectrum Disorder or Parkinson's disease, creating accessibility barriers [74]. Counterfactual data augmentation strategies show promise in reducing spurious correlations by minimally altering sentiment-related features to force causal reasoning [75], but comprehensive debiasing frameworks remain underdeveloped.

Computational complexity impedes deployment in resource-constrained environments. State-of-the-art MLLM-based systems require GPU clusters for inference, rendering them impractical for mobile applications or real-time human-robot interaction [76]. While teacher-student distillation and dynamic attention fusion reduce parameter counts, the accuracy-efficiency trade-off remains severe: distilled models typically sacrifice 3–5 percentage points in accuracy to achieve 10× inference speedup [77]. This gap necessitates architectural innovations beyond simple model compression.

Interpretability deficits hinder trust and clinical adoption. Despite MLLM advances in generating sentiment rationales, most production systems remain black boxes where failure modes are unexplainable [78]. Without transparent attribution of sentiment decisions to specific cross-modal cues, clinicians cannot validate system outputs against professional judgment—a prerequisite for medical deployment [79]. Current explainability methods focus predominantly on textual features while neglecting acoustic and visual contribution analysis.

5. Emerging Trends and Future Directions

Four converging trends define the next phase of MSA research. First, *multimodal foundation models* pretrained on massive heterogeneous datasets (e.g., video-text-audio corpora) are replacing task-specific architectures, enabling zero-shot transfer to low-resource sentiment domains through prompt engineering [80]. Second, *physiological integration* is expanding beyond laboratory settings through wearable sensors and contactless monitoring, enabling continuous emotional assessment in naturalistic environments [81]. Third, *personalized sentiment models* adapting to individual emotional baselines show promise for neurodiverse populations, moving beyond one-size-fits-all approaches that fail atypical expressers [82]. Fourth, *hybrid symbolic-neural architectures* combine neural pattern recognition with structured reasoning to handle culturally-specific irony and subtle social nuances that pure statistical models miss [83].

Critical research gaps requiring immediate attention include:

1. **Robustness to extreme modality degradation**: Current systems fail catastrophically when >50% of a modality's signal is corrupted (e.g., heavy background noise in audio). Research must develop architectures with graceful degradation properties maintaining >70% accuracy under severe signal loss.

2. **Cross-cultural generalization**: Existing benchmarks lack diversity in cultural expression norms. New datasets capturing non-Western emotional display rules (e.g., East Asian restraint versus Mediterranean expressiveness) are essential for globally deployable systems.

3. **Causal reasoning beyond correlation**: Most fusion methods capture statistical associations between modalities without modeling causal emotional mechanisms. Integrating causal inference frameworks could distinguish genuine emotional signals from coincidental correlations (e.g., distinguishing nervous laughter from genuine amusement).

4. **Energy-efficient real-time architectures**: Mobile deployment requires models operating under 1W power budgets with <100ms latency. Current approaches sacrifice too much accuracy for efficiency; novel hardware-aware architectures co-designed with edge processors are needed.

5. **Ethical guardrails for emotional profiling**: As MSA penetrates healthcare and workplace monitoring, technical solutions for privacy preservation (e.g., on-device processing, differential privacy for emotional data) must advance alongside accuracy improvements to prevent manipulative applications.

6. **Longitudinal sentiment dynamics**: Current benchmarks focus on isolated utterances, neglecting how sentiment evolves across extended interactions (e.g., therapy sessions spanning hours). Architectures modelling emotional trajectories rather than static snapshots represent an underexplored frontier.

6. Conclusion

Multimodal Sentiment Analysis has matured from simplistic text-audio-video concatenation into a sophisticated discipline integrating representation learning, cross-modal alignment, and reasoning-capable architectures. The methodological evolution—from classical machine learning through deep learning hybrids to transformer-based fusion and finally MLLM-guided reasoning—reflects a fundamental shift toward systems that approximate human-like sensory integration and emotional inference [84]. Contemporary architectures leveraging contrastive learning and cross-modal attention achieve state-of-the-art performance (86%+ Acc-2 on CMU-MOSI), while MLLM integration introduces unprecedented explain ability through natural language rationales [85].

Nevertheless, significant gaps persist between laboratory benchmarks and real-world deployment requirements. Modality misalignment, missing data robustness, demographic bias, and computational constraints collectively limit practical utility outside controlled environments [86]. The field must pivot from pure accuracy maximization toward holistic system design prioritizing robustness, efficiency, and ethical deployment—particularly as MSA applications expand into sensitive domains like mental healthcare and workplace monitoring [87].

The strongest architectural direction combines the reasoning capabilities of large models with the efficiency of distilled student networks through uncertainty-aware collaborative systems [88]. This hybrid paradigm balances performance ceilings with real-time constraints, offering a viable path toward deployable systems. However, technical advances alone are insufficient; the field requires parallel progress in dataset diversity, causal modeling frameworks, and ethical guardrails to ensure MSA technologies benefit all populations equitably [89].

Looking forward, the convergence of multimodal foundation models, physiological sensing, and personalized adaptation will enable sentiment systems that understand emotional expression not as static snapshots but as dynamic, culturally-situated phenomena unfolding across time [90]. Realizing this vision demands interdisciplinary collaboration between machine learning researchers, affective scientists, ethicists, and domain experts—a synthesis essential for developing artificial emotional intelligence that genuinely enhances human well-being rather than merely optimizing classification metrics.

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